

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4

Total Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct:

I v.surya satvika(192524254) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

*I understand that any violation of _____
academic integrity rules will result in disciplinary action.*

Signature of the student:v.s.satvika

Annexure A

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications*.

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

(a)Article Citation Details

- **Author(s):** Write the complete names of all authors.
- **Title:** Write the complete title of the research article.
- **Journal/Conference:** Mention the name of the journal or conference.
- **Year:** Mention the year in which the article was published.
- **DOI:** Provide the DOI of the research article.
- **Domain/Application Area:** Mention the field or application area in which the research is conducted.
- **ML Problem:** Clearly state the specific machine learning problem addressed by the researchers.

(b)

Article Summary

The selected research article focuses on the application of machine learning to solve a specific problem in the chosen domain. The main objective of the study is to develop an effective machine learning approach that can improve the accuracy, efficiency, or decision-making capability of the existing system. The authors identify a research gap in existing methods and explain the limitations of previously proposed approaches. To address this gap, they propose a suitable machine learning technique and evaluate its performance using an appropriate dataset and experimental methodology. The study compares the proposed approach with existing methods and analyses the obtained results using suitable evaluation metrics. The findings demonstrate the effectiveness of the proposed method for the selected application. Overall, the article contributes to the field by providing an improved machine learning approach and demonstrating its potential for practical applications.

Task 2: Methodology Analysis

- (a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.
- (b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

(a)

ML Technique and Dataset Description

- **Machine Learning Technique Used:** Identify the main machine learning algorithm used in the article.
- **Type of Technique:** State whether it belongs to Inductive Learning, Decision Trees, Statistical Learning Methods, or Reinforcement Learning.
- **Algorithm:** Mention the specific algorithm or variant used.
- **Dataset Size:** State the total number of samples or records in the dataset.
- **Features:** Mention the important input features or variables.
- **Dataset Source:** State where the dataset was obtained.
- **Data Preparation:** Explain the preprocessing steps such as data cleaning, normalization, feature selection, or data splitting.
- **Training and Testing:** Mention how the dataset was divided into training, validation, and testing sets, if provided.

(b)

Mapping the Technique to CO4

The machine learning technique used in the article can be mapped directly to the CO4 topics because CO4 focuses on implementing machine learning techniques for classification and decision-making. If the article uses a Decision Tree or one of its variants, it can be connected to the Decision Tree concept. If it uses Naïve Bayes, Logistic Regression, or SVM, it can be mapped to Statistical Learning Methods. Similarly, if the study uses an agent-based learning approach involving rewards and decision-making, it can be mapped to Reinforcement Learning. The methodology provides a practical application of the concepts studied in CO4.

Methodological Strength:

The methodology uses a systematic machine learning approach and evaluates the model using suitable experimental procedures and performance metrics. This helps demonstrate the effectiveness of the proposed technique.

Methodological Limitation:

The methodology may be limited by the size, quality, diversity, or source of the dataset. The performance obtained from a particular dataset may not always generalize to different real-world conditions.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked. **(b)** Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings? **(c)** Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Key Results

- **Accuracy:** Mention the reported accuracy of the proposed model.
- **Precision:** Mention precision if provided.
- **Recall:** Mention recall if provided.
- **F1-Score:** Mention the F1-score if provided.
- **AUC-ROC:** Mention the AUC-ROC value if provided.
- **Reward/Convergence:** For reinforcement learning articles, mention reward improvement or convergence results.
- **Comparison:** Compare the proposed model with the other models used in the article.
- **Best Model:** Identify which model achieved the best performance.
- **Overall Finding:** State the main result obtained by the authors.

Critical Evaluation

The results reported in the article provide evidence regarding the performance of the proposed machine learning technique. The evaluation metrics used by the authors are appropriate for

measuring model performance; however, high performance values should be interpreted carefully. The results may be influenced by factors such as dataset size, dataset quality, class imbalance, feature selection, and the evaluation procedure. There may also be a possibility of bias if the dataset does not sufficiently represent real-world conditions. To strengthen the findings, the authors could conduct additional experiments using larger and more diverse datasets. Cross-validation, testing on an independent dataset, comparison with additional baseline models, and statistical significance testing could also improve the reliability of the results.

Real-World Applicability

The proposed machine learning technique has potential for deployment in real-world applications related to the domain discussed in the article. It can be used to support automated prediction, classification, monitoring, or decision-making. However, transferring the research model from an experimental environment to an industrial environment may involve several challenges. These include the availability of sufficient high-quality data, computational requirements, scalability, model maintenance, integration with existing systems, privacy concerns, and ethical considerations. Therefore, further testing under real-world conditions is necessary before large-scale deployment.

Task 4: Reflection & Learning Outcome

- (a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.
- (b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

Three Key Insights

1. Understanding Machine Learning Techniques

The first important insight gained from the article is how machine learning techniques can be applied to solve practical problems. This improves the understanding of the machine learning algorithms discussed in CO4.

2. Importance of Data and Evaluation

The second insight is that the quality and preparation of the dataset have a major influence on model performance. Proper evaluation metrics are also necessary to determine whether a model performs effectively.

3. Practical Application of CO4 Concepts

The third insight is that concepts such as Inductive Learning, Decision Trees, Statistical Learning Methods, and Reinforcement Learning can be applied to real-world classification and decision-making problems.

Proposed Novel Experiment or Enhancement

If I were to extend this research, I would conduct an experiment using a larger and more diverse dataset and compare the existing algorithm with an improved machine learning approach. The new dataset would help determine whether the model can generalize effectively to different conditions. Additional evaluation metrics and cross-validation could also be included to obtain more reliable results. This enhancement is feasible because publicly available datasets and modern machine learning tools can be used for experimentation. The expected impact would be improved model reliability, better generalization, and greater potential for real-world deployment.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand